

Spatial Distribution of Bidirectional Charging Stations by Simulation and Evaluation of Traffic and Grid Data

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Submitted: 01 August 2026 | Revised: 25 March 2026 | Accepted: 01 August 2026

Abstract. This papers goal is the investigation on how bidirectional charging stations can be distributed across a dynamic area by combining traffic simulation with partially extracted grid data from official sources. We model charging demand as a spatiotemporal process and evaluate candidate locations with respect to accessibility, traffic flow, and waiting time. The proposed workflow integrates mobility traces, station utilization forecasts, and synthetic grid constraints to identify robust deployment strategies. Results indicate that jointly optimizing traffic and grid criteria can reduce peak grid load while improving charging availability in high-demand districts.

Keywords: Bidirectional charging, traffic simulation, SUMO, OpenStreetMap, spatio-temporal analysis, charging demand, V2G

Nomenclature

BEAM	Behavior, Energy, Autonomy, and Mobility
EV	Electric Vehicle
MATSim	Multi-Agent Transport Simulation
POI	Point of Interest
SoC	State of Charge
SUMO	Simulation of Urban Mobility
TraCI	Traffic Control Interface
V2G	Vehicle-to-Grid
OSM	OpenStreetMap
CS	Charging Station

1. Introduction

The rapid growth of electric mobility is shifting charging-infrastructure planning from static placement rules toward data-driven decision support [1]. Bidirectional charging adds a further dimension, since vehicles are no longer only energy consumers but can also feed power back to the grid, a concept known as Vehicle-to-Grid (V2G), which turns charging points into distributed flexibility assets [2]. Deciding where to place such stations therefore requires understanding where and when charging demand actually concentrates.

Because electric vehicles still represent a small share of the fleet, there is little real-world data on where charging stations should be placed and how intensively they are used. At the same time, uncoordinated charging concentrates electrical load in space and time and can stress local distribution grids [3], which makes the spatial

and temporal structure of demand a central planning question. Existing siting studies, however, often rely on data or tooling that is not openly reproducible [4]. We therefore ask: *how can candidate locations for charging stations be ranked from openly available data, based on their accessibility and on how charging demand is distributed over space and time?*

To address this, we build a simulation workflow on openly available data. Road-network and land-use information are taken from OpenStreetMap (OSM), and traffic is simulated microscopically in Simulation of Urban Mobility (SUMO), controlled through its Python interface Traffic Control Interface (TraCI). From land-use Point of Interests (POIs)—residential buildings, offices, and amenities—we derive travel demand, and we model charging as a *spatiotemporal* process, that is, demand that varies simultaneously by location and time of day. The model distinguishes public charging stations from private home wallboxes with vehicle-specific access control. A custom web interface, which is inspired by SUMO’s web interface, lets users select an area of interest on a map and generate ready-to-run scenarios directly, lowering the barrier to reproducing and extending the analysis. Candidate sites are then assessed in terms of accessibility, station utilization, and the resulting local charging load.

The scope of this study is the planning and evaluation stage. We focus on the spatial and temporal distribution of charging demand and load under realistic mobility patterns. A detailed electrical model of the distribution grid (power flow, feeder and transformer constraints) as well as battery degradation and market or pricing mechanisms are beyond its scope and left to future work.

1.1 Motivation

Urban charging demand is highly uneven across time and location, which can cause queues at popular stations while other sites remain virtually unused. At the same time, cars that are parked for extended periods of time (in residential or commercial areas) can remain plugged in, creating opportunities for bidirectional energy

exchange. The spatial distribution of charging stations therefore affects both user experience and grid load. For example, placing stations in high-traffic areas may improve accessibility but could also lead to congestion and increased waiting times. Conversely, locating stations in less busy areas might reduce queues but could limit their utilization and the potential for vehicle-to-grid (V2G) services. Understanding the spatial and temporal patterns of charging demand is therefore a prerequisite for infrastructure planning. However, there is a lack of open, reproducible methods for assessing candidate locations based on real-world mobility patterns and grid constraints.

1.2 Problem Statement

Given the aforementioned challenges, the problem statement of this paper can be formulated as follows: How can we determine the optimal spatial distribution of bidirectional charging stations, while regarding existing charging stations, in an urban area by integrating traffic simulation data with partially synthetic grid constraints, in order to maximize accessibility and minimize waiting times for electric vehicle users?

1.3 Contributions

This paper contributes a practical and reproducible workflow for planning bidirectional charging infrastructure with coupled traffic and grid evaluation. This work contains a user interface for deterministic scenario-based traffic generation along with a thorough analysis of candidate station portfolios under varying demand and grid conditions. By integrating synthetic grid constraints with mobility-based accessibility metrics, we provide a holistic framework for siting decisions that can be adapted to different urban contexts and policy objectives.

The main contributions of this work are as follows:

- A simulation-based siting pipeline that jointly evaluates mobility accessibility and distribution-grid impact.

- A multi-criteria scoring framework for comparing candidate station portfolios under proto-realistic demand variability.
- A reference implementation in Python to support reproducible experiments and scenario-based sensitivity studies.

2. Related Work

Planning charging infrastructure has been studied extensively, and recent reviews organize the field along the dimensions of station siting, sizing, grid reinforcement, trip-considerations and the treatment of uncertainty [5, 13]. Most contributions optimize either where demand is best served or how the grid copes with the additional load. Bidirectional charging, in which vehicles also feed power back, is only beginning to be integrated into such planning frameworks [9].

2.1 Traffic-Aware Siting

Traffic-aware siting methods estimate charging demand from origin-destination flows, activity patterns, and route choices, often placing stations so as to maximize coverage of traffic along routes. Data-driven analyses suggest that charging demand is largely shaped by proximity to amenities, Electric Vehicle (EV) density, and adjacency to high-traffic corridors, which motivates deriving demand from land-use data as we do [6]. Microscopic traffic simulators are increasingly used for this purpose. SUMO in particular provides built-in energy and charging-infrastructure models and has been used to assign electric vehicles to charging stations from simulated traffic data [7, 8].

A different approach than SUMO, is a mesoscopic traffic simulation model that uses a network of nodes and links to represent the road network, and simulates traffic flow based on traffic demand and road capacity. This approach can be used to estimate charging demand at a more aggregated level, and can be useful for planning charging infrastructure in areas with limited data availability or where detailed traffic simulation is not feasible. Multi-Agent

Transport Simulation (MATSim) is an example of a mesoscopic traffic simulation model that has been used for this purpose, and paired with Behavior, Energy, Autonomy, and Mobility (BEAM) it is possible to simulate the impact of charging infrastructure on a dynamic traffic network [11, 12].

2.2 Grid-Aware Siting

Grid-aware planning approaches prioritize feeder hosting capacity, voltage stability, and peak-load mitigation, frequently relying on metaheuristic optimization to jointly site and size stations under electrical constraints [5]. While they improve technical feasibility, they can yield a poor user experience when travel behavior is not modeled in sufficient detail. The placement of charging stations is often motivated not only by charging demand and by economic advantages, but also by repercussions for the electrical network [14]. While a reasonable amount of studies have been conducted on the siting of charging stations, there are only few that consider working with large-scale electric vehicle charging infrastructure [15].

2.3 Coupled and Bidirectional Approaches

A smaller but growing body of work couples transportation simulation with power-system analysis, for example through co-simulation frameworks that pass simulated charging loads to a distribution-grid model [10]. For bidirectional charging, planning models increasingly distinguish public and private chargers and account for uncertainty in demand and renewable generation [9]. Integrated and openly reproducible frameworks that combine microscopic mobility simulation with bidirectional charging at the level of individual public stations and private home wallboxes, however, remain limited. This paper addresses that gap on the mobility-and-demand side, while leaving a detailed electrical model of the distribution grid to future work.

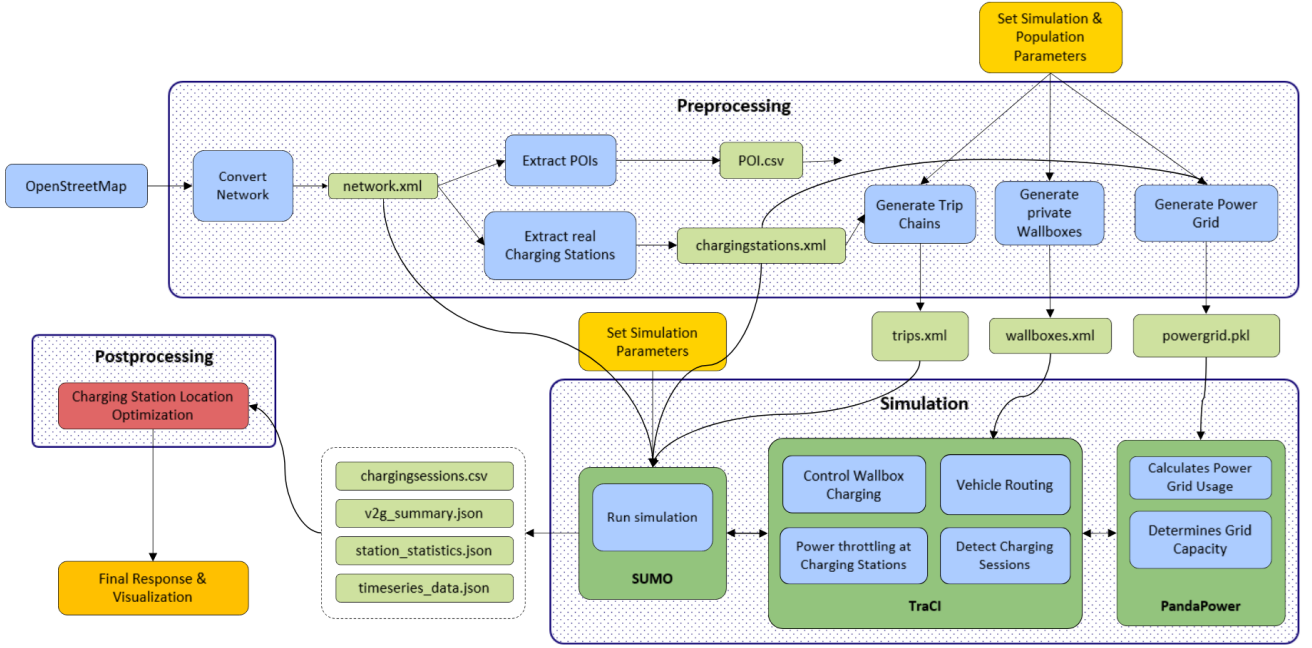


Figure 1: Complete Workflow Pipeline: From OSM data acquisition to scenario generation, execution and manipulation to analysis and visualization.

3. Methodology

SUMO is used as the microscopic traffic simulator and TraCI for real-time control, combined with a grid-impact model and a statistical distribution of home-charging devices, to evaluate the spatial distribution of bidirectional charging stations. Two siting strategies – *Saturate-and-Prune* and *Clustering* – are implemented and detailed in section 3.2.

3.1 System Overview

Figure 1 illustrates the complete workflow pipeline. The framework is organized into four consecutive stages: (i) data acquisition, (ii) preprocessing and scenario generation, (iii) coupled traffic-grid simulation, and (iv) analysis and visualization. Each stage produces intermediate artifacts that are consumed by the subsequent stage, so that the entire process is reproducible from a single set of user-defined parameters.

Data acquisition. The pipeline retrieves information like the road network, electrical grid topology, and candidate POIs from OSM. The road network is converted into a SUMO-

compatible format using the `netconvert` tool, while grid features and POIs are persisted for later preprocessing. This ensures that the simulation has accurate and relevant data to work with, allowing for realistic modeling of traffic flow and charging behavior in the region of interest.

Preprocessing and scenario generation.

The `chargingstations.xml` file contains both, existing and candidate stations produced by the placement algorithm, and is utilized not only for the generation of the synthetic power grid, but also in assistance with the POI file to generate trip chains. Trip chains map individual agents to a sequence of activities and destinations, from which their charging demand and low-State of Charge (SoC) behavior emerge. In parallel, the private-wallbox generator assigns home-charging devices to a share of agents following a statistical distribution, and the power-grid generator couples the charging stations and wallboxes to their nearest buses to form the synthetic distribution grid. To finalize the preprocessing, user-set parameters (EV penetration rate, wallbox share, battery capacities, grid voltage levels) are applied to the mechanisms of the

trip chain, private wallbox, and power grid generation, yielding a fully specified, deterministic scenario.

Coupled traffic-grid simulation. After the preprocessing the files `osm.passenger.trips.xml`, `private_wallboxes.xml`, and `power_grid.pkl` will then be used by SUMO to simulate traffic flow and charging behavior. TraCI, SUMO and PandaPower are the key components of the simulation, which is executed in a time-stepped manner. TraCI allows for real-time control of the simulation, enabling the dynamic adjustment of traffic flow and charging behavior based on an Application Programming Interface (API) via Python. SUMO simulates traffic on a microscopic level (i.e., individual vehicles and their interactions), while PandaPower evaluates the impact of charging behavior on the synthetic distribution grid. While SUMO and TraCI handle the traffic simulation, SUMO and PandaPower do not communicate directly. Instead, TraCI acts as a bridge between the two, allowing for the exchange of information and control signals. This enables the simulation to capture the complex interactions between traffic flow and grid behavior, providing the ability for SUMO to incorporate grid constraints and for PandaPower to calculate the impact of charging behavior on the grid.

Analysis and visualization. The simulation produces a variety of output files, including `station_statistics.json`, `v2g_summary.json`, `charts_data.json`, `charging_sessions.csv` and `model_log_data.csv`. These persisted outputs are analyzed and visualized to evaluate the placement algorithm along metrics such as charging sessions, net energy flow over time, peak grid power, and individual charging station utilization (section 3.3). The results are rendered as plots and maps in the web interface, and inform future iterations of the placement algorithm. Figure 2 shows an example visualization with generated (blue)



Figure 2: Resulting visualization with generated (blue) and extracted (green) CS for example run.

and extracted (green) charging stations for a single run. The heatmap and orange circles indicate approximate areas for charging station placement, while the blue and green dots represent the actual locations of the generated and extracted charging stations, respectively.

3.2 Charging Station Placement Algorithm

Two siting strategies are implemented to compare different placement policies:

- **Saturate-and-Prune:** A naive iterative approach where candidate stations are placed on all viable road segments, ensuring maximum coverage. After each trial execution, charging stations with less than a 5% utilization rate are removed, and the simulation is rerun until all remaining stations exceed this threshold. A 5% threshold was chosen to ensure that stations are neither underutilized nor overburdened, while also allowing for a sufficient number of stations to be placed. This approach is designed to maximize accessibility and coverage, but may result in a higher number of stations being placed than necessary, leading to potential underutilization of some stations.
- **Clustering:** A heuristic that prioritizes locations with high traffic flow, potential traffic congestion, and consideration for vehicles that enter a state of low SoC. Placement follows the traffic flow, aiming to max-

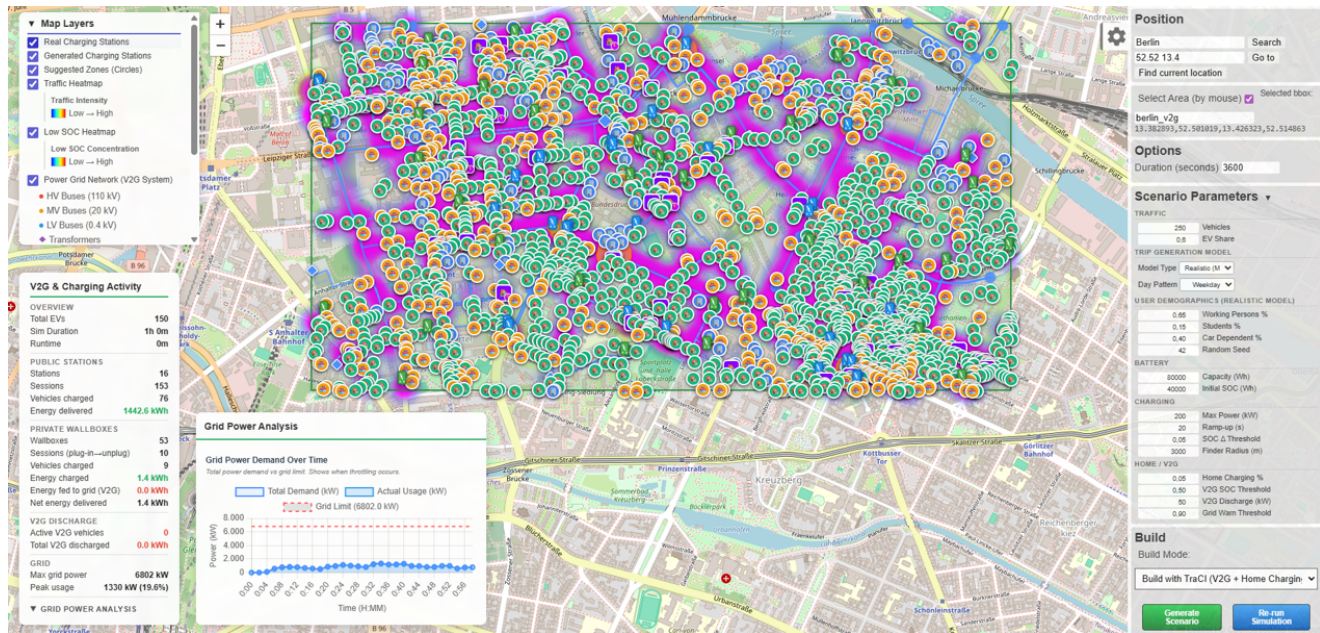


Figure 3: Example run of the workflow: the web interface with all data layers enabled.

imize accessibility where vehicles are most likely to need charging while reducing detours.

3.3 Evaluation Metrics

Mobility accessibility. CS-placement is evaluated based on traffic flow, charging behavior, energy consumption, and SoC. The gathered data is used to calculate the accessibility of the charging stations, which is defined as the percentage of agents that are able to access a charging station within a certain distance or time threshold. Distance and time are key factors in determining the accessibility of charging stations, as they directly impact the convenience and feasibility of using the stations for charging. Distance is an honest measure of how far an agent must travel to reach a charging station, while time takes into account factors such as traffic conditions, congestion, timely availability of charging stations, time and power (kWh) required to charge.

Grid impact. During the simulation, PandaPower solves an AC power flow on the synthetic distribution grid at every step, evaluating bus voltages and the loading of lines and transformers as constraints that bound the power

allocated to each station. The grid impact reported per run is the aggregate demand and the peak grid power relative to the available grid capacity, which captures the stress the charging activity imposes on the grid without exceeding its voltage and loading limits.

Aggregation. The evaluation metrics are aggregated to provide a comprehensive assessment of the charging station placement algorithm. This includes combining mobility accessibility and grid impact metrics to evaluate the overall performance of the charging station network. The aggregation process involves analyzing the trade-offs between accessibility and grid impact, allowing for a balanced evaluation of the placement algorithm's effectiveness.

4. Experiments

4.1 Datasets

The experiment is conducted on mobility data, which is generated by an algorithm that simulates movement patterns of agents in a given area. The algorithm takes into account various factors such as land use, transportation infrastructure and natural behavior of agents to generate realistic mobility data. The generated

data is then used to evaluate the performance of the proposed charging station placement algorithm. Before running the simulation, the dataset consists of surface level mobility (agent start- and endpoints, OSM-data), which is used for traffic flow, POIs and electrical grid information. When a simulation is run, the dataset is enriched with additional information such as charging behavior, energy consumption, specific locations of agents and their SoC. The datasets are generated using a combination of real-world data and synthetic data, ensuring that the results are both realistic and generalizable to other urban areas. The datasets are also designed to be scalable, allowing for the evaluation of the placement algorithms on larger areas with more agents and charging stations.

4.2 Experimental Setup

The reference scenario covers a one kilometer by one kilometer area in Berlin, Germany. It is populated with 250 agents that are distributed across the area's residential zones and assigned activities and destinations from the generated trip chains. All agents are modelled as electric vehicles with a battery capacity of 80 kWh, initialised at 50 % SoC. The simulation covers one hour of simulated time with a time step of 10 s, which keeps the coupled traffic and power-flow computation tractable while still resolving individual charging sessions. Charging power per station is capped at 200 kW with a 20 s ramp-up, V2G discharge is enabled above 50 % SoC with up to 50 kW per vehicle, and 5 % of the agents own a private wallbox. The complete parameter set is persisted per run in `sim_params.json`, so that every reported figure can be traced back to the configuration that produced it.

The results reported in section 4.3 are obtained from a single run with random seed 42 on the infrastructure that is already present in the area, that is, the 26 charging stations extracted from OSM. This run serves two purposes: it quantifies the accessibility gap of the existing infrastructure, and it provides the charging-demand log from which the Clustering strategy derives its candidate sites. Because a single run cannot separate systematic effects from the

noise that dynamic routing introduces, the numbers below are reported as observations of one scenario instance rather than as statistically significant differences between the two placement strategies. A multi-seed comparison of Saturate-and-Prune against Clustering along the metrics of section 3.3 is the subject of ongoing work and is discussed in section 5.

4.3 Results

Accessibility of the existing infrastructure.

Over the simulated hour, the fleet's mean SoC drops from 50.0 % to 33.9 %, with a large spread across agents (std. 19.6 %, median 43.4 %). The tail of that distribution is the relevant part. In it, 28.0 % of all agents end the hour below 20 % SoC, 22.8 % drop below 10 % at some point during the run, and 3.2 % are fully depleted. At the same time, only 6 of the 250 agents (2.4 %) manage to charge at all, and they do so at only 4 of the 26 available stations. The remaining 22 stations record no session, because they are not reachable along the routes the agents actually drive. The four used stations are occupied for 232 to 319 of the 360 simulation steps, that is, between 64 % and 89 % of the simulated hour, delivering 149.6 kWh in total at an effective power of 47.5 kW. This is the failure mode the placement algorithms are meant to address. The network is not short of stations in aggregate, but short of stations where and when demand occurs, so a small subset saturates while most of the fleet is never served. Table 1 summarises the run.

Derived charging demand. The DBSCAN stage of the Clustering strategy condenses the 2249 logged low-SoC samples into 21 spatially coherent demand clusters. A sample is one vehicle at one 10 s time step below 30 % SoC, so the sample count measures dwell time rather than demand, and a single vehicle standing for 27 minutes already contributes 163 samples. The demand of a cluster is therefore derived from the vehicles themselves. Samples inside a cluster are grouped per vehicle and split into separate visits whenever a vehicle is absent for more than

Table 1: Baseline run on the existing OSM infrastructure: 250 agents, one hour of simulated time, 10s time step, seed 42.

Metric	Value
Agents (all EV)	250
Logged vehicle states	60 049
Mean SoC, start $\rightarrow$ end	50.0 $\rightarrow$ 33.9 %
Std. of end SoC	19.6 %
Agents below 20 % SoC at end	28.0 %
Agents below 10 % SoC (any time)	22.8 %
Agents fully depleted	3.2 %
CS available (OSM)	26
CS used	4
Agents that charged	6 (2.4 %)
Occupancy of used CS	64–89 %
Energy delivered	149.6 kWh
Effective charging power	47.5 kW
Low-SoC samples logged	2249
Demand clusters (DBSCAN)	21
Distinct vehicle visits	128
Energy demand of clusters	6393 kWh
Charge points estimated	184
CS sites generated	29

five minutes, and each visit is charged with the energy it would need to reach 80 % SoC. Across all clusters this yields 128 visits by 90 vehicles and an energy demand of 6393 kWh. Table 2 lists the ten clusters with the highest demand. The distribution is strongly skewed. The two largest clusters carry 55 % of all visits and 54 % of the energy demand, while 13 of the 21 clusters consist of a single visit by a single vehicle. The mean SoC inside a cluster ranges from 5.5 % to 30.0 % (mean 18.7 %), which separates clusters that mark genuine range emergencies from those that merely mark opportunistic charging locations. Cluster radii stay compact at 25.0 m to 66.6 m (mean 31.1 m), so each cluster is small enough to be served by a single station site rather than requiring a distributed set of sites. Two of the 21 clusters lie 3.3 m apart and describe the same demand twice. Discounting the duplicate, the non-redundant demand is 4680 kWh.

Grid-aware siting. All 21 clusters fall within reach of the synthetic distribution grid, at a mean distance of 105 m to the nearest connec-

Table 2: The ten charging-demand clusters of the baseline run with the highest energy demand. *Vis.* is the number of distinct vehicle visits, *Veh.* the number of vehicles behind them, *kWh* the energy those visits need to reach 80 % SoC, *CP* the resulting number of charge points and *Cap.* the grid capacity at the nearest connection point.

ID	Vis. (#)	Veh. (#)	SoC (%)	Rad. (m)	kWh	CP (#)	Cap. (kW)
30	35	24	19.8	30.7	1712.5	46	5000
29	35	17	10.2	25.0	1712.5	46	5000
5	18	16	14.3	66.6	952.2	26	5000
16	10	8	13.7	52.0	531.6	15	5000
9	10	6	19.3	31.1	497.6	14	5000
39	3	2	21.9	37.9	156.1	5	500
25	2	2	7.1	29.8	102.9	3	5000
36	2	2	29.5	54.1	100.8	3	500
11	1	1	5.5	25.0	59.6	2	5000
34	1	1	8.7	25.0	57.0	2	500

All 21 clusters: 128 visits by 90 vehicles, 6392.6 kWh, 184 charge points; SoC 18.7 ± 8.2 %; radius 31.1 ± 11.8 m; grid rating 9 excellent, 12 good.

tion point (max. 249 m). Nine clusters are rated *excellent* and twelve *good* with respect to available grid capacity; none had to be discarded for lack of a feasible connection. Four clusters are limited to a 500 kW connection while the remaining seventeen have 5000 kW available. A charge point is a single connector at which one vehicle can charge, so a station site aggregates several of them; the heuristic sizes a cluster as its hourly energy demand divided by the throughput of one 50 kW charge point at 75 % utilisation, and then caps the result at what the local grid connection can supply. For the 21 retained clusters the grid cap is not binding, since the largest cluster requires 46 charge points against the 100 its 5000 kW connection would allow. It did bind for three further clusters that were subsequently discarded for being too close to an existing station. The demand estimate over all clusters is 184 charge points, distributed over the 29 station sites the strategy generates, i.e. about six connectors per site. In this scenario the limiting factor is neither the number of sites nor the grid connection, but

the fact that demand concentrates on very few locations, as table 2 shows directly.

Interpretation. The run supports the premise of the approach but does not yet discriminate between the two placement strategies. The observed concentration of demand, with over half of all visits in two clusters and 85 % of the existing stations unused, is precisely the situation in which a traffic- and SoC-driven heuristic is expected to outperform uniform saturation. Confirming that expectation requires the multi-seed comparison described in section 5. The numbers reported here establish the baseline against which it will be measured.

5. Discussion

5.1 Practical Example

5.2 Limitations

6. Conclusion

This paper presented a simulation-driven method for distributing bidirectional charging stations under joint traffic and grid criteria. The evaluation demonstrates that coordinated planning can improve both charging accessibility and electrical reliability.

6.1 Future Work

Future work will include stochastic optimization under forecast uncertainty and integration of dynamic pricing signals. We also plan to validate the framework with larger real-world datasets and pilot deployments in mixed urban-suburban regions.

Acknowledgments

The authors thank the Fraunhofer Center of Application KEIM for supporting this research work.

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